Big Data System Design: Case Studies

Course: **MBAI 5110G-Big Data System Design**
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Case Study: Netflix

Netflix operates a fully digital streaming platform serving over 150 million global subscribers, generating massive volumes of real-time user interaction data. Every search, click, pause, rating, and viewing session creates structured and unstructured data that feeds into its Big Data ecosystem. Using advanced analytics, machine learning, and AI-driven recommendation systems, Netflix builds detailed subscriber profiles to understand individual viewing behavior, preferences, and engagement patterns. This data-driven approach enables the platform to personalize content recommendations, which account for nearly 80% of viewer activity.

To support this personalization on a scale, Netflix relies on a distributed Big Data architecture capable of handling high volume, velocity, and variety of data. Real-time data ingestion pipelines, scalable cloud storage, and machine learning processing layers work together to generate customized recommendations instantly. The recommendation engine not only enhances user experience but also improves customer retention, contributing nearly $1 billion in value, while reducing marketing and advertising costs. By predicting what audiences are likely to watch, Netflix strategically invests in content production and maintains a competitive advantage in the streaming industry.

Questions

1. Identify the types of data (structured and unstructured) generated by Netflix and explain how each supports personalization.
2. Design a high-level Big Data system architecture for Netflix, clearly explaining the data source, ingestion, storage, processing, and analytics layers.
3. How does Netflix address the challenges of the 5Vs of Big Data (Volume, Velocity, Variety, Veracity, Value) in its Big Data environment?
4. What machine learning techniques could be used in Netflix's recommendation system? Explain their role in system design.
5. If you were designing Netflix's Big Data infrastructure, how would you ensure scalability, fault tolerance, and data security?